

An Overview of Open Source **Cloud** Platforms *for* Enterprises

This article covers the characteristics of the open source cloud and the open source based cloud computing layers, to help enterprises make the right choice.

An open source cloud is any cloud service or solution that is developed using open source technologies and software. This covers any public, private or hybrid cloud models providing SaaS, IaaS and PaaS that have been built and operate entirely on open source technologies.

In today's transformational digital journey, the business, social, economic and technology trends play a major part in shaping the future of an enterprise. Cloud computing has become central to many enterprise IT models, and a number of enterprise architects are trying to make cloud systems as effective and beneficial as possible.

Industry adoption of cloud platforms

IT spending on cloud platforms is projected to reach 60 per cent of all IT infrastructure, and 60-70 per cent of all software services and technology spending by 2020. Forbes forecasts that 83 per cent of enterprise workloads will be in the

cloud by 2020 [<https://www.idc.com/getdoc.jsp?containerId=US43171317>]. Subscription-based Software-as-a-Service (SaaS) is also projected to grow at a CAGR of 18 per cent by 2020 [<https://www.forbes.com/sites/forbestechcouncil/2018/07/05/four-trends-in-cloud-computing-cios-should-prepare-for-in-2019/#69125c2c4dc2>].

The growth rate for investments in Platform-as-a-Service (PaaS) will almost double from 32 per cent in 2016 to 56 per cent in 2019, making it the fastest-growing sector of cloud platforms [https://www.pointivity.com/latest-technology_trends].

As per Gartner, the Infrastructure-as-a-service (IaaS) market is predicted to reach US\$ 72.4 billion worldwide by 2020 [<https://www.gartner.com/newsroom/id/3871416>].

Of the 83 per cent of enterprise workloads that will be in the cloud by 2020, 41 per cent will run on public

